

2.5 → Expenses + profit (Cont...)

2.5.1 → UW Expenses: Exposure/Policy-based method

→ An alternative to premium-based projection method is to use a method that projects fixed expenses based on exposures or # of policies, i.e. the exposure/policy-based projection method.

→ Under this method, variable expenses are handled the same way as in other methods, i.e. divided by premiums. However, historical fixed expenses are divided by historical exposures or policy counts, rather than premium. The choice depends on whether fixed expenses are assumed to be constant (or each exposure / policy count, divide historical expenses by exposures) or for each policy (in which case, divide by the # of policies).

→ The historical expense ratios for each category use the following data:

Expense	Data Used	Fixed Divided By	Variable Divided By
General Acquisition	Countrywide	Earned Exposure	Earned Premium
Other Acquisition	Countrywide	Written Exposure	Written Premium
Commissions and Brokerage	Countrywide/State	Written Exposure	Written Premium
Taxes, Licenses, and Fees	State	Written Exposure	Written Premium

→ The steps for determining the UW expense provision under the Exposure/Policy-based Projection method are:

- 1) Split each expense into fixed & variable components
- 2) Derive the expense ratio for each year & category. Divide by exposures (or policy counts) for fixed expenses & premium for variable expenses
- 3) Select a ratio for each expense type
- 4) Sum the fixed expense ratio selections for each expense category to find the total fixed expense provision. Repeat for variable expenses

→ The procedure results in the average fixed expense per exposure. So if the projected fixed expense ratio is needed instead, it can be found as

$$\text{Projected Fixed Expense Ratio} = \frac{\text{Avg Projected Fixed Expense per Exposure}}{\text{Projected Avg Premium}}$$

→ Potential distortions (which can be remedied slightly (not within here though))

→ While the Exposure/policy-based projection method corrects the distortions caused by the other two methods, it still has its own shortcomings. These are discussed below, along w/ potential adjustments that may be needed

- Expense categorization into fixed & variable components often relies on judgement.
- C/F fixed expenses are allocated to states based on exposure or policy distributions but actual average fixed expense levels may differ by location [e.g., higher advertising costs in some states]
- Some "fixed" expenses can vary by a certain characteristic (e.g. higher advertising costs for new business). This can result in inequitable allocations of fixed expenses; i.e. the rate distribution significantly changes or varies by state.
- When there are similar values for the projected average expense per exposure, this implies that expenses are changing proportionately to exposures. However, not all fixed expenses behave this way due to economies of scale.

→ Trending

- variable expenses don't require trending as they are a constant percentage of premium. However, fixed expenses will likely need to be trended, as they are a constant dollar amount
- For an inflation-sensitive exposure base (e.g. payroll per \$1M), trending is not necessary if expenses & exposures are trending at the same rate
- For a non-inflation-sensitive exposure base (e.g. car year or house year) or for policy counts, the average fixed expenses are expected to change over time, & thus trending is needed.
- selecting a trend for expenses → Internal expense data can be used, but can be volatile ⇒ can use government indices along w/ incorporating any anticipated changes in internal company practices.

→ Find trend for expenses the same way as for premium/benefit. The trend period will be from the average date expenses were incurred in the historical experience period to the average date they will be incurred in the period rates are expected to be in effect.

→ for expenses incurred at the beginning of a policy, trend using the average date that policies are written.

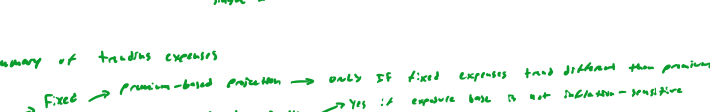
→ for expenses incurred throughout a policy, trend using the average date that policies are earned (i.e. half a policy period longer than)

→ One possible way to simplify expense trending is to assume that all expenses are incurred in the same way (i.e. either all at inception or all throughout the policy period). The impact of this simplification will depend on the magnitude of the expense trend & the portion of premium that fixed expenses represent.

→ Another possible simplification is to treat historical fixed expenses from a single date, rather than determining the trend length for each historical year separately. For instance, assume a 3-year average expense ratio is selected. This ratio could be trended from the midpoint of the 3-year period. This should give approximately the same result as trending each year's expense ratio separately & then averaging the results. If the ratio selected was only the latest year, there would be no difference.

$$\text{average to single} \Rightarrow \text{trend} \approx \text{trend this individually} \Rightarrow \text{average trended this to the} \Rightarrow$$

→ Summary of trending expenses



→ The methods for deriving expense provisions

$$\text{Premium-based method} \Rightarrow \text{Fixed} = \frac{\text{Total expenses} \times \% \text{ of fixed expenses}}{\text{WP} \text{ or EP}}$$

$$\text{Variable} = \frac{\text{Total expenses} \times \% \text{ of variable expense}}{\text{WP} \text{ or EP}}$$

$$\text{Exposure-based projection} \Rightarrow \text{Fixed} = \frac{\text{Total expenses} \times \% \text{ of fixed expense}}{\text{WP or EP} \downarrow \text{exposure}}$$

2.5.2 → UW Profit Provision

→ Total profit = Investment income + UW profit

→ Using the fundamental insurance equation → UW profit = Premium-bills - LAE - UW Expenses

→ Permissible loss ratio (PLR)

→ The variable permissible loss ratio (VPLR) is the percentage of each premium dollar that is intended (or "permitted") to pay for projected losses + LAE & projected fixed expenses. The remaining portion of each premium dollar is then intended to pay for variable expenses & profit. The VPLR is calculated as

$$\text{VPLR} = 1 - \text{Variable expense \%} - \text{Target UW profit \%}$$

$$\downarrow = 1 - V - Q_F$$

→ The total permissible loss ratio (PLR) is the percentage of each premium dollar that is intended (or "permitted") to pay for projected losses + LAE. The remaining portion is intended for all UW expenses & profit. The total PLR is calculated as:

$$\text{PLR} = 1 - \text{Total expense \%} - \text{Target UW profit \%}$$

$$\downarrow = 1 - F - V - Q_F$$

$$\downarrow = 1 - (F + V + Q_F)$$

↳ Where F is the fixed expense ratio

→ VPLR & PLR are the same when all expenses are treated as variable expenses.

$$\text{profitability} = \text{margin} \text{ or } \text{margin} / \text{loss} = \%$$

→ if state a concept as a system, explain the concept as well

★ B.1) layout the margin & choose different ratios

→ How these different ratios? ⇒ Higher premium for higher rates & lower premium for lower rates

2.6 → Overall rate level indicators

2.6.1 → Pure premium method

→ Pure premium method → Indicates avg rate =
$$\frac{\text{Pure premium including LAE} + \text{Fixed UW expense per exposure}}{1 - \text{Variable UW expense \%} - \text{Target UW profit \%}}$$

$$P_F = \frac{L + E_L + E_F}{1 - V - Q_F} \Rightarrow \text{VPLR}$$

Fixed expense per is just $\frac{\text{Projected fixed expenses}}{\# \text{ policies}}$

$$\text{per exposure: } \bar{P}_F = \frac{L + E_L + E_F}{1 - V - Q_F}$$

→ If developments & true-trend trends basis & other adjustments

$$\text{loss} + \text{LAE} = \text{non-cat loss} + \text{LAE} \times \text{LDR} \times \text{VPLR factor} \times \text{Current trend factor} \times \text{Projected trend factor}$$

⇒ by exposures to get PP

→ For order of operations → Develop basis, then trend

→ When selecting method → Sometimes you just have to read the problem & choose an interpretation that makes sense.

If you're worried that your assumption isn't correct, just state what your assumption is & move forward anyway. Unless your assumption is illegal, that's pretty much all credit

2.6.2 → Loss ratio method

→ Loss ratio method → Indicates rate change =
$$\frac{\text{Loss} + \text{LAE ratio} + \text{Fixed expense ratio}}{1 - \text{Variable UW expense \%} - \text{Target UW profit \%}} - 1$$

→ Derivation → Indicated change factor =
$$\frac{P_F}{P_C} \Rightarrow \text{From PP method} \Rightarrow \text{Premium} + \text{LAE}$$

$$= \frac{L + E_L + E_F}{1 - V - Q_F} + \frac{E_F}{P_C}$$

$$= \frac{\text{loss} + \text{LAE ratio} + F}{1 - V - Q_F} \Rightarrow \text{Fixed expense ratio}$$

$$\Rightarrow \text{Indicated avg rate change} = \text{Indicated change factor} - 1$$

→ Problems solving → order doesn't matter when loading / calculating VPLR into losses

→ Can select loss from all years & add trend loss VPLR

or Can load LAE to all years & then select loss + LAE ratio

→ When comparing loss ratios for averaging, company needs to be accounted for

⇒ i.e. OLEP, developed basis, same trended

⇒ ensures apples to apples comparisons across years / classes/etc.

→ Selecting final LR

→ use straight-average if stable / assume developments are just random fluctuations

→ use weighted-average if results are more volatile

→ PP method preferred when can't get OLEP, e.g. if historical rate change is far from 1 that's available

→ Comparison of methods

→ There are two main differences between the pure premium method (PP method) & the LR method (LR method).

★ all → The underlying loss measure used in the calculations is different. The PP method utilizes pure premium & the LR method utilizes the loss ratio. This means that the premium of C/F is needed for the LR method, but not for the PP method. In addition, clearly defined expenses are needed for the PP method, but not for the LR method.

→ PR of this, the PP method is typically used when premium at the C/F is not available or difficult to calculate accurately. For instance, if there are significant changes to rating variables for an insurance product w/ a complex rating algorithm, then can make it difficult to calculate actual premiums

→ on the other hand, the LR method is preferred when exposure data is not available or exposures are not clearly defined for an insurance product. For example, commercial general liability policies can have several sub-lines to cover a wide variety of risks, but these sub-lines may have different exposure bases, which can make it difficult to establish a consistent exposure measure.

→ 2) → The two methods result in different outputs. The PP method results in an indicated rate, while the LR method results in an indicated rate change.

→ Since the LR method involves calculating the percent change from current rates, this method cannot be used for a new company that wants to determine the indicated rate. In that case, external data or judgement can be used to estimate expected PP & expense provisions & to select a target profit provision to use in the PP method.

→ Converting between indicated average rate & indicated avg rate change

$$\Rightarrow \frac{\text{ind rate}}{\text{current rate}} - 1 = \text{ind rate change} \downarrow$$

$$(1 + \text{ind rate change}) \times \text{Avg current rate} = \text{ind avg rate}$$

→ results from PP method vs LR method should be equivalent

(assuming consistent data & assumptions) ↳ B.2) using multiplication method for use & extension of exposures for another

→ Stillwater →

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Year	Current	Current	Current	Current	Current	Current	Current	Current	Current	Current
2020-20	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029

$$(11) \text{ Any weighted partial loss + LAE ratio} = \frac{E(10) \times (9)}{\text{year}}$$

$$(12) \text{ Creditability} = (12)$$

$$(13) \text{ Complement of creditability} = (13)$$

$$(14) \text{ Creditability-weighted trended} = (12) \times (13) + \text{weighted average loss + LAE ratio} = (12) \times (13) + (13)$$

$$(15) \text{ Loaded cat loss} = (15)$$

$$(16) \text{ Margin for experience} = (16)$$

$$(17) \text{ Total prospective loss + LAE ratio} = (14) + (15) + (16)$$

$$(18) \text{ Permissible loss + LAE ratio} = (17) \Rightarrow 1 - \text{VPLR}$$

$$(19) \text{ Indicated rate level needed} = \frac{(17) / (18) - 1}{(19)}$$

→ Calculating final LR

→ can do weighted average
$$\frac{\text{Sum}(E \times \text{loss})}{\text{Sum}(E \times \text{EP})}$$
 OR straight average
$$\frac{1}{n} \sum \frac{\text{loss}}{\text{EP}}$$

Mathematically, this is the same as doing a traditional weighted average (weighted by denominator)

→ When working w/ pure premium (loss ratio), can work w/ losses & convert to PP at end (i.e. by dividing by exposure)

OR can work w/ PP from beginning → need w/ basis though

↳ trend, low adjustments, etc.

→ factors that affect loss cat trends → This is!! Stillwater vs ISO

→ risk of business, inflation, selection deductions/policy limits, etc.